

CUET

PHYSICS

REVISION AND SHORT NOTES



- EXAM FOCUSED
- CONCISE CONCEPTS
- IMPORTANT FORMULAS

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TARGET CUET 2026

+ SHORTCUTS & TIPS

CUET Fiziks Prep

Designed for CUET UG, JEE, and NEET Aspirants

About the Author

Radheshyam is a dedicated Physics educator and founder of *CUET Fiziks Prep*, a platform committed to guiding students towards success in competitive examinations such as CUET, JEE, and NEET. With a strong academic background and a passion for teaching, he specializes in simplifying complex physics concepts into clear, concise, and exam-oriented content.

He holds a **Master of Science (M.Sc.) in Physics** from *Jamia Millia Islamia, Delhi*, and a **Bachelor of Science (B.Sc.) in Physics** from *Hindu College, University of Delhi*. His academic journey reflects a deep understanding of core physics principles combined with practical problem-solving skills.

Through his teaching methodology, Radheshyam focuses on conceptual clarity, strategic preparation, and effective revision techniques. His notes are designed to be concise, formula-driven, and highly relevant for examinations, helping students maximize their performance in limited time.

He continues to mentor thousands of aspirants through digital platforms, aiming to make quality education accessible and impactful for every learner.

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Chapter 1

Electrostatics

1. Electric Charge Fundamentals

Electric charges are the intrinsic property of matter that gives rise to electric and magnetic interactions.

1.1 Types and Nature

- **Polarity:** Two types exist—Positive (+) and Negative (−).
- **Basic Rule:** Like charges repel; unlike charges attract.
- **Material Behavior:**
 - *Conductors:* Allow free movement of charges (Metals, Human body, Earth).
 - *Insulators:* Resist charge movement (Glass, Plastic, Wood).

Fundamental Properties of Charge

1. **Additivity:** Total charge of a system is the simple algebraic sum.
2. **Conservation:** Charge cannot be created or destroyed; it only transfers between bodies.
3. **Quantization:** All free charges are integral multiples of the elementary charge e .

$$q = ne \quad \text{where } e = 1.602 \times 10^{-19} \text{ C}$$

2. Coulomb's Law

The mutual electrostatic force between two point charges (q_1, q_2) is directly proportional to the product of charges and inversely proportional to the square of the distance (r).

Vector Form of Coulomb's Law

$$\mathbf{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}_{21}$$

- **Permittivity (ϵ_0):** $8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$
- **Proportionality Constant (k):** $\approx 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$

3. The Electric Field (**E**)

An electric field is the region around a charge where its influence can be felt by other charges.

3.1 Field Strength

The field at a distance r from a point charge q :

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Note: For a system of charges, use the **Principle of Superposition**:

$$\mathbf{E}_{net} = \mathbf{E}_1 + \mathbf{E}_2 + \cdots + \mathbf{E}_n$$

3.2 Electric Field Lines

Visual representations of the field:

- Start at $+$ and end at $-$.
- Density of lines $\propto$ Magnitude of field.
- **Never cross** each other.
- Do not form **closed loops**.

4. Electric Dipole

A system of two equal and opposite charges separated by a small distance $2a$.

Dipole Moment (**p**)

$$\mathbf{p} = q \times (2\mathbf{a})$$

Direction: Always from Negative ($-$) to Positive ($+$).

4.1 Electric Field of a Dipole

For a short dipole where $r \gg a$:

1. **Axial Point:** $E = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$ (Direction along **p**)
2. **Equatorial Point:** $E = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$ (Direction opposite to **p**)

4.2 Dipole in a Uniform External Field

- **Net Force:** Always zero.
- **Torque (τ):** $\tau = \mathbf{p} \times \mathbf{E} = pE \sin \theta$

5. Electric Flux (Φ)

Flux measures the number of electric field lines passing through a given area.

$$\Phi = \int \mathbf{E} \cdot d\mathbf{S} = ES \cos \theta$$

6. Gauss's Law

Total flux through any closed Gaussian surface is $1/\epsilon_0$ times the net charge enclosed.

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

7. Major Applications of Gauss's Law

7.1 Linear Charge Distribution (Infinite Wire)

Consider a wire with linear charge density λ :

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

7.2 Surface Charge Distribution (Infinite Sheet)

- **Non-conducting Sheet:** $E = \frac{\sigma}{2\epsilon_0}$
- **Conducting Plate:** $E = \frac{\sigma}{\epsilon_0}$

8. Spherical Charge Distributions

This is the most critical section for CUET entrance exams.

8.1 Hollow Conducting Sphere / Shell

Total charge Q resides on the surface of radius R .

- **Outside ($r > R$):** $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
- **On Surface ($r = R$):** $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$
- **Inside ($r < R$):** $E = 0$

8.2 Solid Non-Conducting Sphere (Uniformly Charged)

Total charge Q is distributed throughout the volume of radius R .

- **Outside ($r \geq R$):** $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
- **Inside ($r < R$):** $E = \frac{1}{4\pi\epsilon_0} \frac{Qr}{R^3}$ or $E = \frac{\rho r}{3\epsilon_0}$

Quick Comparison Table

Configuration	Inside ($r < R$)	Outside ($r > R$)
Hollow Shell	$E = 0$	$E \propto 1/r^2$
Solid Sphere	$E \propto r$	$E \propto 1/r^2$

9. Summary of Relationships (MCQ High-Yield)

- **Point Charge:** $E \propto r^{-2}$
- **Dipole:** $E \propto r^{-3}$
- **Infinite Line Charge:** $E \propto r^{-1}$
- **Infinite Sheet:** $E \propto r^0$ (Constant)

10. Practice Strategy for CUET UG

- Always check if the question mentions a **conducting** or **non-conducting** material.
- Remember that **Electric Field Lines** never exist inside a conductor in electrostatic equilibrium.
- **E** is always perpendicular to an equipotential surface (covered in Chapter 2).

Chapter 2

Electrostatic Potential Capacitance

1. Electrostatic Potential and Work

- **Conservative Nature:** Electrostatic force is conservative; work done depends only on initial and final positions, not the path.
- **Potential (V):** Work done per unit positive charge in bringing it from infinity to a point.

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

- **Superposition:** For multiple charges, $V_{net} = V_1 + V_2 + \dots + V_n$.

Potential of a Dipole

At a distance r from the center of a dipole $\mathbf{p}$:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$$

Special Cases:

- *On Axis ($\theta = 0^\circ$):* $V = \frac{kp}{r^2}$
- *On Equatorial Line ($\theta = 90^\circ$):* $V = 0$

2. Equipotential Surfaces

A surface where the potential is constant at every point.

- **Work Done:** Moving a charge on an equipotential surface requires **zero work**.
- **Direction of $\mathbf{E}$:** The electric field is always **perpendicular** to the equipotential surface.
- **Field Gradient:** $\mathbf{E} = -\frac{dV}{dr}$ (Field points toward decreasing potential).

3. Potential Energy of a System

- **Two Charges:** $U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$
- **Dipole in Uniform Field:** $U = -\mathbf{p} \cdot \mathbf{E} = -pE \cos \theta$

4. Capacitors and Capacitance

Capacitance (C) measures the ability of a system to store charge.

$$C = \frac{Q}{V} \quad (\text{Unit: Farad, F})$$

4.1 Parallel Plate Capacitor

For plates of area A separated by distance d :

$$C = \frac{\epsilon_0 A}{d}$$

- **With Dielectric (K):** $C_{new} = KC_0 = \frac{K\epsilon_0 A}{d}$
- **Electric Field inside:** $E = \frac{\sigma}{\epsilon_0}$ (or $\frac{\sigma}{K\epsilon_0}$ with dielectric).

5. Combinations of Capacitors

Series Combination	Parallel Combination
$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$	$C_{eq} = C_1 + C_2 + \dots$
Charge (Q) is same for all.	Potential (V) is same for all.

6. Energy Stored in a Capacitor

Energy (U) stored in the electric field between plates:

$$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}\frac{Q^2}{C}$$

Energy Density (u): Energy per unit volume $u = \frac{1}{2}\epsilon_0 E^2$.

7. High-Yield Extras for CUET UG

- **Dielectric Breakdown:** The maximum electric field a material can withstand without becoming a conductor.
- **Common Potential (V_c):** When two capacitors are connected:

$$V_c = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

- **Van de Graaff Generator:** Uses a moving belt to build up high potential (millions of volts) on a hollow spherical shell. Used for accelerating particles.

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Chapter 3

Current Electricity

1. Electric Current and Resistance

- **Electric Current (I):** The net charge passing through a cross-sectional area per unit time.
- **Electromotive Force (emf):** The work done per unit charge by a source to move charge from lower to higher potential. It is the voltage difference in an *open circuit*.

Ohm's Law

The current I flowing through a conductor is proportional to the voltage V across its ends ($V \propto I$).

$$V = RI$$

Resistance (R): Depends on length (l) and area (A):

$$R = \frac{\rho l}{A}$$

where ρ is the **resistivity** of the material.

2. Drift Velocity and Resistivity

- **Current Density ($\mathbf{j}$):** Amount of charge flowing per second per unit area normal to flow.

$$\mathbf{j} = nq\mathbf{v}_d$$

- **Drift Velocity ($\mathbf{v}_d$):** The average velocity gained by electrons due to an electric field.

$$v_d = a\tau = \frac{eE\tau}{m}$$

where τ is the **relaxation time** (average time between collisions).

Formula for Resistivity

$$\rho = \frac{m}{ne^2\tau}$$

Note: ρ increases linearly with temperature for metals.

3. Circuit Components and Combinations

3.1 Internal Resistance

When a source of emf ε is connected to an external resistance R , the terminal voltage V is:

$$V = IR = \frac{\varepsilon}{R + r}R$$

where r is the **internal resistance** of the source.

3.2 Resistance Combinations

Series Combination	Parallel Combination
$R_{eq} = R_1 + R_2 + \cdots + R_n$	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$
Current remains same.	Potential difference remains same.

4. Kirchhoff's Rules

Essential for complex circuit analysis:

1. **Junction Rule (KCL):** At any junction, $\sum I_{in} = \sum I_{out}$. (Based on *Conservation of Charge*).
2. **Loop Rule (KVL):** The algebraic sum of changes in potential around any closed loop is zero ($\sum \Delta V = 0$). (Based on *Conservation of Energy*).

5. Measuring Instruments

- **Wheatstone Bridge:** An arrangement of four resistors. At the **null-point** (no current through galvanometer):

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

- **Potentiometer:** A device to compare emfs or measure internal resistance. It works on the condition of **no current flow**, making it more accurate than a voltmeter.